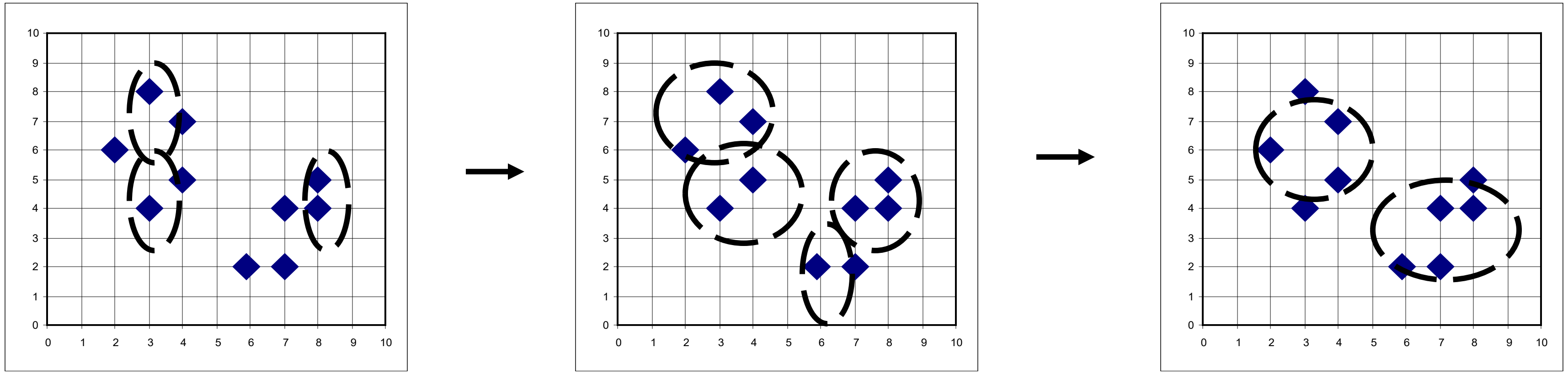
Agglomerative Clustering Algorithm

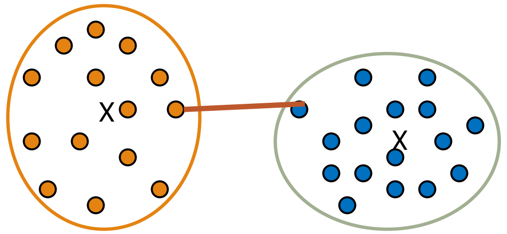
- AGNES (AGglomerative NESting) (Kaufmann and Rousseeuw, 1990)
  - Use the single-link method and the dissimilarity matrix
  - Continuously merge nodes that have the least dissimilarity
  - Eventually all nodes belong to the same cluster
- Agglomerative clustering varies on different similarity measures among clusters
  - Single link (nearest neighbor)
  - Complete link (diameter)
  - Average link (group average)
  - Centroid link (centroid similarity)

Agglomerative Clustering Algorithm

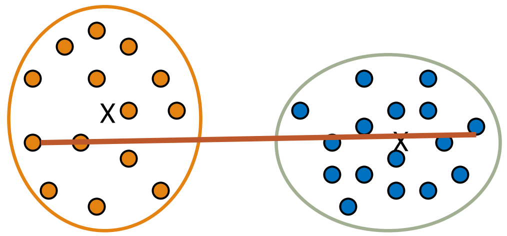


Single Link vs. Complete Link in Hierarchical Clustering

- Single link (nearest neighbor)
  - The similarity between two clusters is the similarity between their most similar (nearest neighbor) members
  - Local similarity-based: Emphasizing more on close regions, ignoring the overall structure of the cluster
  - Capable of clustering non-elliptical shaped group of objects
  - Sensitive to noise and outliers

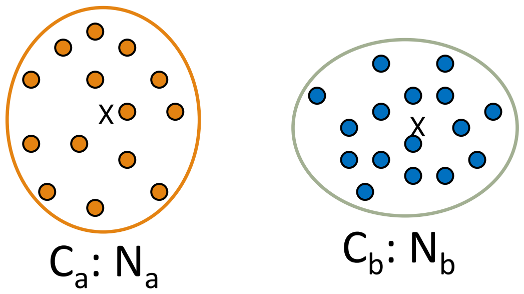


- Complete link (diameter)
  - The similarity between two clusters is the similarity between their most dissimilar members
  - Merge two clusters to form one with the smallest diameter
  - Nonlocal in behavior, obtaining compact shaped clusters
  - Sensitive to outliers

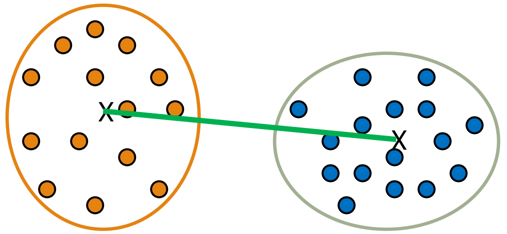


Agglomerative Clustering: Average vs. Centroid Links

- Agglomerative clustering with average link
  - Average link: The average distance between an element in one cluster and an element in the other (i.e., all pairs in two clusters)
    - Expensive to compute



- Agglomerative clustering with centroid link
  - Centroid link: The distance between the centroids of two clusters



- Group Averaged Agglomerative Clustering (GAAC)
  - Let two clusters C_a and C_b be merged into $C_a \cup C_b$
  - The new centroid is $c_{a \cup b} = \frac{N_a c_a + N_b c_b}{N_a + N_b}$, where N_a and c_a are the cardinality and centroid of cluster C_a, respectively
  - The similarity measure for GAAC is the average of their distances

Agglomerative Clustering with Ward's Criterion

- Suppose two disjoint clusters C_i and C_j are merged, and m_{ij} is the mean of the new cluster
- Ward's criterion: $W(C_i, C_j) = \frac{n_i n_j}{n_i + n_j} \left| m_i - m_j \right|^2$